

What every symbol on slide 7 means

Ramp-constrained multi-period economic dispatch • companion sheet to the seminar deck

1. The program, for reference

$$\min_{p, f, s} \underbrace{\sum_{t \in \mathcal{T}} \sum_{g \in \mathcal{G}} c_g p_{g,t} \Delta t}_{\text{fuel bill}} + \underbrace{\sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{N}} \text{VOLL} \cdot s_{i,t} \Delta t}_{\text{penalty for unserved load}}$$

subject to

$$\sum_{g \in \mathcal{G}_i} p_{g,t} - (D_{i,t} - s_{i,t}) = \sum_{\ell: \text{fr}(\ell)=i} f_{\ell,t} - \sum_{\ell: \text{to}(\ell)=i} f_{\ell,t} \quad \forall i \in \mathcal{N}, \forall t \in \mathcal{T} \quad (\text{C1})$$

$$\underline{P}_g \leq p_{g,t} \leq \overline{P}_g \quad \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \quad (\text{C2})$$

$$-\overline{F}_\ell \leq f_{\ell,t} \leq \overline{F}_\ell \quad \forall \ell \in \mathcal{L}, \forall t \in \mathcal{T} \quad (\text{C3})$$

$$p_{g,t} - p_{g,t-1} \leq \text{RU}_g \quad \forall g \in \mathcal{G}, t = 2, \dots, T \quad (\text{C4})$$

$$p_{g,t-1} - p_{g,t} \leq \text{RD}_g \quad \forall g \in \mathcal{G}, t = 2, \dots, T \quad (\text{C5})$$

$$p_{g,1} - P_g^0 \leq \text{RU}_g, \quad P_g^0 - p_{g,1} \leq \text{RD}_g \quad \forall g \in \mathcal{G} \quad (\text{C6})$$

$$p_{g,t} \geq 0, \quad 0 \leq s_{i,t} \leq D_{i,t} \quad (\text{C7})$$

C4, C5 and C6 are the only lines that are new. Everything else is the single-period transport model with a t added to every subscript.

2. Sets and indices — the filing cabinet

Symbol	Name	What it counts	In our case
$i, j \in \mathcal{N}$	buses (nodes)	Junction boxes — where wires meet and load hangs	$ \mathcal{N} = 3$
$g \in \mathcal{G}$	generators	The taps you can open and close	$ \mathcal{G} = 2$
$\mathcal{G}_i \subseteq \mathcal{G}$	generators at bus i	Which taps are bolted to <i>this</i> junction box	$\mathcal{G}_1 = \{1\},$ $\mathcal{G}_2 = \{2\}$
$\ell \in \mathcal{L}$	lines (branches)	The pipes between junction boxes	$ \mathcal{L} = 3$
$t \in \mathcal{T}$	time periods	Frames of the filmstrip, $\mathcal{T} = \{1, \dots, T\}$	$T = 24$
$\text{fr}(\ell)$	from-bus of ℓ	The end the pipe's arrow is painted <i>from</i>	—
$\text{to}(\ell)$	to-bus of ℓ	The end it points <i>to</i>	—

On fr and to. These are a *sign convention*, not a claim about direction. They say which way we agree to call positive. If the solver returns $f_{\ell,t} < 0$, the power is genuinely flowing the other way, and that is a legitimate answer. (The formulation writes $\text{fr}(\ell)$ and $\text{to}(\ell)$ rather than $f(\ell)$ and $t(\ell)$ precisely so the to-bus never collides with the time index t .)

3. Parameters — the numbers stamped on the equipment

These are **given**. The optimiser reads them and cannot change them.

Symbol	Name	Units	Physically	In our case
c_g	marginal cost	\$/MWh	The price tag on the fuel feeding tap g	G1: 10 G2: 30
$\overline{P}_g$	maximum output	MW	How far open the tap physically goes	both: 250
$\underline{P}_g$	minimum output	MW	How far <i>closed</i> it goes while still running	both: 0
RU_g	ramp-up limit	MW/h	How many turns of the hand-wheel you may make <i>upward</i> in one period	G1: 30 G2: 250
RD_g	ramp-down limit	MW/h	The same wheel, turning downward	G1: 30 G2: 250
P_g^0	initial output	MW	Where the tap was set the hour <i>before</i> the horizon starts	G1: 100 G2: 50
$\overline{F}_\ell$	thermal rating	MW	The burst rating of pipe ℓ	1–2: 250 1–3: 150 2–3: 250
$D_{i,t}$	demand	MW	How much is drained out of bus i during hour t	peak 364.0 at $t=18$
Δt	period length	h	How long one frame lasts	1
VOLL	value of lost load	\$/MWh	The fine for failing to deliver	5,000

Why VOLL is enormous. At \$5,000/MWh, shedding load is never economic — the solver will burn the most expensive generator in the fleet first. It is a *safety valve*, not a choice: it converts an unhelpful INFEASIBLE into an answer that tells you exactly *where* and *when* the system broke.

4. Decision variables — the knobs the optimiser may turn

Symbol	Name	Range	What you are physically setting
$p_{g,t}$	generation	≥ 0	How far open tap g is during hour t
$f_{\ell,t}$	line flow	free sign	How much power moves through pipe ℓ ; positive means from $\text{fr}(\ell)$ toward $\text{to}(\ell)$
$s_{i,t}$	load shed	$0 \leq s_{i,t} \leq D_{i,t}$	Load at bus i deliberately <i>not</i> served in hour t

Together these form a table with $|\mathcal{G}|$ rows and T columns for p (here $2 \times 24 = 48$ numbers), plus $|\mathcal{L}| \times T$ flows and $|\mathcal{N}| \times T$ shed variables. The static model solved a single column of that table.

5. Notation conventions

Mark	Reads as
$\overline{x}$	An <i>upper</i> bound on x — a ceiling ($\overline{P}_g, \overline{F}_\ell$)
$\underline{x}$	A <i>lower</i> bound on x — a floor ($\underline{P}_g$)
x^0	The value of x <i>before</i> the horizon begins (P_g^0)
$x_{g,t}$	Two indices: the first says <i>which machine</i> , the second <i>which hour</i>
$\sum_{g \in \mathcal{G}_i}$	Add up over every generator sitting at bus i
$\sum_{\ell: \text{fr}(\ell)=i}$	Add up over every line whose from-end is bus i
$\forall$	“for all” — this line is not one constraint but one <i>per</i> listed combination
$\in$	“is a member of”

The $\forall$ multiplies the work. C1 is not a single equation: written $\forall i \in \mathcal{N}, \forall t \in \mathcal{T}$ it stands for $3 \times 24 = 72$ separate equations. C4 and C5 together stand for $2|\mathcal{G}|(T-1) = 92$.

6. Reading each line aloud

	In words
obj	Minimise the fuel burnt across the whole horizon, plus a large penalty for any load left unserved.
C1	At every bus, in every hour: what the generators there inject, minus what the load there actually takes, equals the net power leaving on the wires. Kirchhoff's current law.
C2	No generator runs above its maximum or below its minimum, in any hour.
C3	No line carries more than its rating, in either direction, in any hour.
C4	From one hour to the next, a generator may rise by at most RU_g megawatts.
C5	And it may fall by at most RD_g megawatts.
C6	The first hour is measured against where the machine already was at midnight — the horizon does not start from a blank screen.
C7	Generators do not consume power, and you cannot shed load that was never there.

7. The one structural point

C1, C2, C3 and C7 all carry the index t , but each of them acts *within* a single hour: delete C4–C6 and the problem separates into T independent single-period problems that could be solved one at a time and stapled together.

C4 and C5 are the only constraints that mention *two different values of t at once*. That is what welds the hours into a single program, and it is the entire source of every interesting behaviour in the model — the anticipative dispatch, the prices that leave the range $[\min_g c_g, \max_g c_g]$, and the fact that a fleet with ample capacity can still fail because demand rose too *steeply*.

Full derivation and the DC-OPF extension (C8, C9): docs/DYNAMIC_DISPATCH_FORMULATION.md. Implementation: OPT simple case/OPT part 4 - dynamic ramping.jl. Parameter values above are those actually used in that script.